

Ariel Marie Virgoe Jacobs

Personal Information

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Education

2021– Dec 2026 (expected)	PhD in Atmospheric and Climate Science , <i>University of Washington</i> , Seattle, Washington, Advisor: Dr. Cecilia M. Bitz. Dissertation: Investigating Arctic Cyclones and the Arctic Boundary Layer in Nudged and Free-Running Community Atmosphere Model (CAM) Simulations General Exam passed: September 29, 2025
2021–2024	Master of Science in Atmospheric Sciences , <i>University of Washington</i> , Seattle, Washington, Advisor: Dr. Cecilia M. Bitz. Master's Thesis: Investigating Summer 2022 Arctic Cyclone Structure and Evolution
2017–2021	Bachelor of Science in Meteorology, cum Laude , <i>University of Oklahoma</i> , Norman, Oklahoma, Minors in Mathematics and Computer Science. Honors Thesis: Validation of The North American Regional Reanalysis (NARR) Model Against Radiosonde Profiles in Norman, Oklahoma Advisor: Dr. Phillip Chilson

Professional Experience

Sept 2021–Present	Graduate Research Assistant , <i>University of Washington</i> , Seattle, Washington. My research focuses on Arctic cyclone structure and evolution. I am also interested in the Arctic boundary layer more broadly. Participated in CPEX-AW and CPEX-CV field campaigns.
May 2025–August 2025	Significant Opportunities in Atmospheric Research and Science protégé , <i>University Corporation for Atmospheric Research</i> , Boulder, Colorado. Discovered a low liquid water path bias in Winter 2019-2020 in the Community Atmosphere Model version 7 compared to observations from the recent MOSAiC field campaign. Altering the microphysics was not enough to address the problem as the boundary layer physics was also contributing to the bias.
May 2021–July 2021	Significant Opportunities in Atmospheric Research and Science protégé , <i>University Corporation for Atmospheric Research</i> , Virtual. Utilized multiple datasets including dropsondes, radar data, and novel coyote UAS data to observe the oceanic cold wake and the ABL during Hurricane Maria (2017). Discovered limited dropsondes in Hurricane Maria confirmed previous modeling and observational results for the thermodynamic hurricane boundary layer (HBL) and dynamic HBL heights, and coyote UAS provided useful information in time and space.
Jun 2019–May 2021	Undergraduate Research Assistant , <i>Center for Autonomous Sensing and Sampling</i> , University of Oklahoma. Most recent research project focused on analyzing North American Regional Reanalysis (NARR) data to establish climates for potential Unmanned Aerial Systems (UAS) sites across Oklahoma. Assisted in the maintenance of post-processing software for UAS flight data.
Mar 2018–Nov 2018	Development Operations Intern , <i>DTN (formally Weather Decision Technologies, Inc.)</i> , Norman, Oklahoma. Created scripts and templates for cloud-based development in Amazon Web Services and designed, managed, and maintained tools to automate operational processes. Additionally, built tools for deployment, monitoring and operations and collaborated with developers on software and system optimizations.
Jul 2017–Mar 2018	Meteorological Support Specialist , <i>DTN (formally Weather Decision Technologies, Inc.)</i> , Norman, Oklahoma. Provided tech support for mobile app users and corporate clients regarding issues from subscription renewal to portal set-up for new clients. Used support tools, problem-solving skills, and communication skills to assist customers via the email ticketing system and phone calls.

Awards, Honors, and Certifications

- 2026 Graubard Fellowship in the UW Program on Climate Change (Autumn 2026)
- 2023–Present National Defense Science and Engineering Graduate Fellow
- 2025 DCMIP colloquium participant (self-funded)
- 2023 NCAR ASP colloquium participant
- 2023 NSF GRFP Honorable Mention
- 2023 Distinguished Service to the Department
- 2021 Top Scholar Award, UW Graduate School Fund for Excellence and Innovation
- 2021 Jacobs et al. 2020 selected as an Editor's Choice article in Remote Sensing
- 2020–2021 Kandi & Mark McCasland Award for Outstanding Undergraduate Research Paper
- 2020–2021, 2025 NSF Significant Opportunities in Atmospheric Research and Science Protégé
Accepted in 2020, internship postponed until Summer 2021 due to the pandemic
- 2018 Best Novice (first Hackathon) Hack and placed within the Top Ten teams, Hacklahoma, University of Oklahoma
- 2018 Honors College First Year Research Experience (FYRE) Participant
- 2017–2019 American Meteorological Society Freshman Scholarship Recipient
- 2017–2021 University of Oklahoma Academic Excellence Scholar

Affiliations

- 2025–2026 Identity, Belonging and Inquiry in Science (IBIS) Program Mentor
- 2023–2024 Diversity and Inclusion Group (DIG) Coordinator
- 2022–2024 UW College of Environment Student Advisory Council
- 2020–2021 Atmospheric & Geographic Sciences Diversity & Inclusively Council
- 2020–2021 & 2022–2023 Letters to a Pre-Scientist Pen-pal
- 2017–2021 Alpha Sigma Kappa - Women in Technical Studies
Scholarship Chair Fall 2020, Scholarship Committee Fall 2018
- 2014–Present American Meteorological Society

Refereed Articles and Select Presentations

Published

- 2020 **Jacobs, A.M.**; Bell, T.M.; Greene, B.R.; Chilson, P.B. The Effect of Climatological Variables on Future UAS-Based Atmospheric Profiling in the Lower Atmosphere. *Remote Sens.* **2020**, 12, 2947.

In Review

- 2026 **Jacobs, A.M.**; Blanchard-Wrigglesworth, E.; Bitz, C.M. Investigating Recent Summer Arctic Cyclone Structure and Evolution. *J. Atmos. Sci.* **2026**
- 2026 **Jacobs, A.M.**; Bacmeister, J.T.; Bitz, C.M. Evaluating Model Biases in the Arctic Boundary Layer: Insights from Comparing CAM to the MOSAiC field campaign. *Geophys. Res. Lett.* **2026**

Select Presentations

- 2025 **Jacobs, A.M.**; Blanchard-Wrigglesworth, E., Bitz, C.M.: Investigating Recent Arctic Cyclone Structure, Evolution, and Boundary Layer in the Community Atmosphere Model including sensitivity to the New CLUBB-MF Parameterization. Poster Presentation: American Geophysical Union 2025 Meeting, Washington, DC, December 16th, 2025.
- 2025 **Jacobs, A.M.**; Bacmeister, J.T., Bitz, C.M.: Identifying Model Biases in the Arctic Boundary Layer: Insights from Comparing CAM to the MOSAiC field campaign. Oral Presentation: American Geophysical Union 2025 Meeting, Washington, DC, December 16th, 2025.
- 2024 **Jacobs, A.M.**; Blanchard-Wrigglesworth, E., Bitz, C.M.: Investigating Recent Arctic Cyclone Structure and Evolution. Poster Presentation: American Geophysical Union 2024 Meeting, Washington, DC, December 12th, 2024.

- 2022 **Jacobs, A.M.**; Rozoff, C.; Hawbecker, P.; Chen, S.S., Cione, J.: Observations of Oceanic Cold Wake and the Atmospheric Boundary Layer during Hurricane Maria (2017). Oral Presentation: 102nd Annual Conference of the American Meteorological Society, Virtual, January 24th, 2022.
- 2021 **Jacobs, A.M.**; Bell, T.M.; Greene, B.R.; Chilson, P.B.: The Effect of Climatological Variables on Future UAS-Based Atmospheric Profiling in the Lower Atmosphere. Oral Presentation: 101st Annual Conference of the American Meteorological Society, Virtual, January 12, 2021.

Computer skills

Programming Languages	PYTHON, JAVA, C++, MATLAB, BASH, R, HTML, NODE.JS, $\LaTeX$
Models	CESM, WRF
Cloud Computing Platforms	Amazon Web Services, Google Cloud Platform, Microsoft Azure
Other Applications	GitHub, ArcSight, Maltego, Tableau, Zendesk